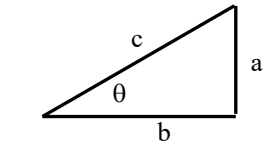
PHYSICS 1101.200 Group Practice Problem 5 October 13, 2026

This is a closed book and closed notes group problem with no connection to any source of information. Calculators are permitted only to do mathematical calculations. No devices that can contact the internet are allowed. The **ONLY** physics formulas that may be used are those given below. Define all symbols and justify all mathematical expressions used. Make sure to state all assumptions & approximations used to solve a problem. If this were a quiz, partial credit would be given for a well-communicated problem-solving framework based on correct physics. Each problem is worth 25 points: **In the context of a unified solution**, credit will be awarded as follows: a useful sketch that shows all relevant quantities, listing the known quantities, and writing the quantity to be determined is 7 points; writing the physics principles useful to solve the problem, writing any assumptions or approximations needed, drawing useful physics diagrams, and writing the equations that apply the physics principles in terms of the quantities from your sketch is 6 points; planning the solution by describing how to use those physics equations to construct an algebraic answer is 5 points; solving those equations for a symbolic result is 3 points; checking the units & reasonableness of the symbolic result is 2 points; calculating a numerical result & stating why its units are correct and giving a reason why that result is not unreasonable is 2 points. If no numerical answer is required, 1 point will be added the planning the solution and 1 point to solving the equations and checking the units & reasonableness of the solution.

Useful Mathematical Relationships:

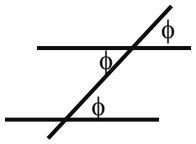


For a right triangle: $\sin\theta = \frac{a}{c}$, $\cos\theta = \frac{b}{c}$, $\tan\theta = \frac{a}{b}$,

$a^2 + b^2 = c^2$, $\sin^2\theta + \cos^2\theta = 1$

For a circle: $C = 2\pi R$, $A = \pi R^2$

For a sphere: $V = \frac{4}{3}\pi R^3$, $S = 4\pi R^2$



If $Ax^2 + Bx + C = 0$, $x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$, $(a+b)^2 = a^2 + 2ab + b^2$

Fundamental Concepts, Principles, and Definitions:

$\sum \vec{F} = m\vec{a}$	$\vec{F}_{A\text{ on }B} = -\vec{F}_{B\text{ on }A}$	$a_{x\text{ average}} = \frac{\Delta v_x}{\Delta t}$	$v_{x\text{ average}} = \frac{\Delta x}{\Delta t}$	$\tau = rF_{\perp}$

Under Certain Conditions:

$F = mg$	$F = \mu N$	$F_{max} = \mu N$	$F = k\Delta L$	$\sum \vec{F} = 0$	$\sum \vec{\tau} = 0$

Useful constants: 1 mile = 5280 ft, g = 9.8 m/s^2 = 32 ft/s^2

Your team has been tasked with investigating incidents of worker back problems that could be caused by routine activities at a shipping facility. Before starting this study, you decide to determine how the force on the spine at the fifth lumbar vertebra, where the spine is connected to the pelvis, depends on the weight of a box lifted improperly from the floor. You have observed that occasionally a person bends over so their back is horizontal and lifts the box straight up. To get a rough idea of the situation, you decide to represent the spine as a uniform horizontal rod with a known length and negligible weight. You also neglect the weight of the back. One end of the rod is where the shoulders support the weight of the box as it is lifted straight up off the floor. The other end of the rod is where the spine is attached to its pivot at the fifth lumbar vertebra. Although the back muscles are complicated, for this simple model you assume that the principal muscle lifting the spine is attached two thirds of the way up the spine at an angle of 2 degrees to the horizontal spine.